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INTRODUCTION



Women, work, and the digital economy

Uma Rani , Ruth Castel-Branco , Shivani Satija  and Mahima Nayar 

Introduction

Digital technologies are bringing about a transformation in the world of work at a rapid pace. Today, digitalisation has penetrated almost all major sectors of the economy (ILO 2021). Digital labour encompasses a wide range of occupations, from software developers and programmers to domestic workers on digital labour platforms, to market vendors and microentrepreneurs who use digital tools to reach customers. The COVID-19 pandemic has further accelerated the process of digitalisation, as work and livelihoods shifted online. Yet it also exposed and exacerbated inequalities between the global North and South, and along the lines of gender, race, caste, and class. Limited access to digital infrastructure, low digital literacy, and repressive sociocultural norms are some of the common reasons cited for this digital divide.

Across the global South, governments have embraced digitalisation with the hope that it will increase productivity and competitiveness and generate jobs. The latter is particularly important given the high and rising unemployment, amidst an ongoing process of deagrarianisation and deindustrialisation. Whereas the structural adjustment policies of the 1980s and 1990s effectively reduced public investment in technological innovation, cutting much of the global South out of the dot-com boom, this new wave of digitalisation seemingly offers an opportunity to make up for the lost decades of neoliberalism. Indeed, digitalisation has been widely promoted by international development agencies as a policy pathway towards sustainable, inclusive, and equitable economic growth, with the ‘potential to improve social and economic outcomes for women (UN Women 2020, 1). However, Huws (2014) argues that digital innovation has facilitated the concentration of capital across industries and geographies, increasing their monopoly power, in a context where states’ regulatory capacity already hangs in the balance. As surplus value is increasingly derived from value extraction rather than commodity production, workers’ bargaining power has been severely undermined.

Over the past decade, platform work has attracted significant interest from scholars in advanced economies of the global North, and it is now gaining attention from scholars in the global South. Much of the contemporary debate emphasises the misclassification of platform workers, lack of state regulation, and challenges in organising workers on digital labour platforms. The focus of the literature has generally been on male-dominated

sectors like e-hailing and delivery work. The collection of papers in the November 2022 special issue of *Gender & Development*, titled ‘Women, Work, and the Digital Economy’, makes two important contributions to the existing scholarship on platform work. First, the issue focuses on women workers – both in women- and male-dominated professions – teasing out the similarities and differences across sectors and countries, predominantly in the global South. Second, the papers explore how women navigate and assert themselves, highlighting the emerging forms of organisation, from WhatsApp groups to co-operatives, to trade unions. By focusing on women workers, the impressive collection of inter-disciplinary papers in this issue locate the debate at the intersection of exploitation, exclusion, and oppression, thereby contributing to the incipient male-dominated literature on emerging forms of worker struggle.

The issue draws empirical evidence from a diverse set of countries from Argentina, Brazil, China, Ghana, India, Kenya, Mexico, Nigeria, Pakistan, South Africa, Spain, Sri Lanka, and Uganda. The papers explore: What has been the impact of digital technologies on the labour process and conditions of work? How have women and other marginalised groups resisted inequities at an individual and collective level? How can states contribute to a sustainable, inclusive, and equitable digital economy through public policy? Many papers adopt a sectoral approach, focusing on beauty work, domestic and care work, food delivery, ride-hailing, factory and home-based work, micro- and small-scale entrepreneurship, and public provisioning. Some explore broader questions of structural transformation and the role of global governance institutions, including international development agencies, in shaping the terms of development. By employing an intersectional lens, the papers and essays examine the multiple vulnerabilities, risks, and precarities – as well as the individual and collective strategies deployed in resisting and reshaping the terms of platform capitalism.

Impact of new technologies on women

There has been significant debate about how new technologies will influence the structure of the labour market, and some analysts have predicted massive jobs losses – at least in the short term (Frey and Osborne 2017). However, there is no consensus in the literature about the potential impacts on jobs and the estimates vary substantially depending upon the country (Grimshaw and Rani 2021). Questions are also being raised about whether an entire occupation is prone to automation or rather a specific task within an occupation (Arntz *et al.* 2019). This is further explored in this issue by von Dietrich and Garcia) in ‘New challenges for women workers in Brazil facing the wave of Industry 4.0 technologies’, who examine 20 main occupations for women in Brazil. They argue that the impact of digital innovations on labour-replacement will vary by occupation and task, but that women are likely to be in more vulnerable occupations due to gender inequalities in the labour market, including their under-representation in STEM (Science, Technology, Engineering, and Mathematics) areas. Nevertheless, there are several mediating factors. For instance, automation is expected to be higher in several low-wage occupations. However, precisely because they are already low wage, they may not necessarily attract

investment in labour-saving technologies. Furthermore, occupations such as care services which are also low paid have the lowest probability of automation. Ultimately, the process of automation is neither predetermined nor linear, and social dialogue can play an important role in determining what technology is adopted, how, and for what end.

Gender digital divide and empowerment

While gender inequalities in the labour market – participation, wages, and education in STEM areas – continue to persist, digital technologies are increasingly seen as a mechanism for a socioeconomic development trajectory and women’s economic empowerment in developing countries. In this context, addressing the gender digital divide is argued to be critical as it offers mobile technologies and digital financial services to achieve gender equality as well as sustainable and inclusive economic growth (OECD 2017, 2018). In addition, the dominant economics discourse suggests that narrowing the digital divide would lead to an increase in women’s labour force participation which would improve their social and economic well-being (UN Women 2020). Several other international institutions such as the World Economic Forum (2021) and the World Bank (2021) in their reports also stress that narrowing the digital divide would be important for addressing gender inequality. While undoubtedly addressing the gender digital divide is crucial, at the core is the question of who gets access to digital networks and technologies and whether it enables women to access employment opportunities and increase incomes so that they can overcome longstanding inequalities.

Dixit and Banday, in their paper ‘Problematising the digital gender gap: invoking decoloniality and intersectionality for inclusive policymaking’, problematise the issue of the digital gender divide and argue that ensuring digital inclusion alone will not automatically lead to gender empowerment and equality. This is largely because the discourse of the ‘gender gap’ is often seen as a ‘problem of rights and economic development’ (p. 437) which are then addressed through neoliberal rationality of entrepreneurship and capabilities, ultimately leading to women’s empowerment. This narrative according to them completely ignores the intersectionality of gendered power relations and contextual factors, and in the process reproduces colonial frameworks of modernity and progress. They further contend that having digital access, skills, or literacy does not automatically transform into empowerment, as there are significant barriers to utilising the economic opportunities due to race or caste, religion, class, and other social inequalities. For instance, women marginalised by caste, culture, and religion, as argued by Majid and Mustafa, might find it difficult to access digital resources or to find jobs even when they have digital skills.

In addition, many women are trapped in traditional family roles and patriarchal norms, which restrict them from accessing and using digital technologies for their own social and economic empowerment. Majid and Mustafa address this issue in their paper, ‘Transformative digital spaces? Investigating women’s digital mobilities in Pakistan’, drawing on in-depth interviews conducted with the domestic, factory, and home-based workers in Pakistan. Their analysis shows the barriers women face in

patriarchal societies in accessing mobile technologies and how they navigate this space and use their mobile phones. The use of mobile phones differs across the different types of work. For instance, home-based workers had a higher incidence of mobile phone ownership than domestic workers and factory workers, which they used to expand their networks and enhance their income. One would have assumed that factory workers would have had greater access to mobile phones, as they travel outside their homes and work in male-dominated spaces. However, access to mobile phones was found to be limited given the prevalence of patriarchal norms and ideologies that seek to discipline women who work outside the home through the 'stigma' attached to such work and the fear of engaging in 'inappropriate behaviour'. The paper also brings out the multiple barriers women face in realising women's economic empowerment in patriarchal societies. Nevertheless, women are consciously trying to circumvent these constraints and policy can play an important role in narrowing the gender divide.

The potential of digital technologies in augmenting informal workers' income is also analysed by de Silva in 'Navigating the digital informal economy during the COVID-19 pandemic: vignettes of Sri Lankan micro- and small-scale entrepreneurs'. She draws on surveys conducted during the COVID-19 pandemic in Sri Lanka to explore how women used digital tools such as social media to run their small enterprises amidst constraints on mobility, concerns over health and safety, and competing care responsibilities. She highlights how social media provided women entrepreneurs with an opportunity to network and expand their clientele, source orders and make payments, and collaborate and build solidarity with other micro and small enterprises. However, while access to social media provided opportunities for women entrepreneurs in urban areas, accessing such technologies remains a challenge for those in other informal markets and in rural areas.

Borborah and Das, in 'Building agency: women vendors and gendered technology in informal markets in Assam', show that women vendors in urban informal markets in India lack access to such digital technologies and mobile phones, and use traditional mechanisms to conduct their business and earn incomes. They argue that the lack of access to mobile phones for women is due to both economic and normative barriers, and power relations within the household. As women's work is often considered to be secondary within the household, scarce resources such as mobile phones are often given to men. The paper highlights that when the use of mobile technologies is constrained due to family responsibilities and societal norms, women build alternative social networks with their customers to overcome the challenges of the digital divide.

While accessing and using digital technologies remains a challenge for many, we find that governance is becoming increasingly digitalised in a move towards increasing accessibility, availability, and efficiency. Raj and Juned, in 'Gendered identities and digital inequalities: an exploration of the lived realities of the transgender community in the Indian digital welfare state', explore the impact of public digital systems on the everyday lives of transgender people, who belong to marginalised populations. They show that lack of access to digital infrastructure and digital literacy becomes a challenge for navigating public digital systems, especially for marginalised women and transgender people. These communities are subjected to arbitrary rules and additional requirements to prove their

identity, thus increasing alienation from public provisioning. This paper further argues that while access is restricted at multiple levels, digitalisation adds another layer of complexity for transgender communities in India who already face multiple challenges and inequalities.

Digital labour platforms: empowerment or dis(empowerment)?

There is an ongoing debate about the possibilities of digitalisation in increasing the labour force participation rates and providing opportunities for marginalised women and groups to access labour markets. This contention is often based on the notion that platforms provide women with the flexibility to balance paid work with household responsibilities. Anwar, in his paper 'Platforms of inequality: gender dynamics of digital labour in Africa', challenges the global policy narrative that stresses that the platform economy will 'increase women's participation in the labour markets and framed as a panacea to poverty and inequality among women' (p. 748).

Several papers in this special issue question the notion that platforms provide flexibility and that participation on these platforms will inevitably lead to improved working conditions for women. Women's narratives in this issue, irrespective of sector, reveal that their decision to join the platforms was motivated by the flexibility and autonomy it offered in terms of working hours. Tandon and Sekharan, in 'Labouring (on) the app: agency and organisation of work in the platform economy', highlight that for both married women with children and for single mothers, the flexibility was appealing as it allowed them to 'balance' paid work with household responsibilities. Kalla, in her narrative 'Hacking platform capitalism: the case of domestic workers on South Africa's SweepSouth platform', also shows that some young workers look at platforms as an alternative for temporary livelihoods and that it exposes them to varied types of clients. However, the flexibility propagated by the platform is ultimately a myth as women often have to work longer hours and at odd hours during the day. As Centeno Maya *et al.* in 'Food delivery workers in Mexico City, a gender perspective on the gig economy' observe, sometimes delivery workers end up working six days a week for more than six hours each day to earn enough income. Flexibility often becomes a tool for legitimising double shifts for 'women' (workplace and household work), while there is a constant struggle between the production and the social reproduction process which deepens the vulnerability of women platform workers, and they rarely enjoy the flexibility as discussed by Kwan in 'Gendered precarious employment in China's gig economy: exploring women gig drivers' intersectional vulnerabilities and resistances'.

The misclassification of workers on platforms as 'self-employed' or 'independent contractors' or 'partners' allows platforms to circumvent labour laws and regulations, further marginalising domestic workers who are often Black migrant women and already vulnerable (Sibiya and du Toit in 'Sweeping up decent work: paid domestic work and digital platforms in South Africa'). Kalla, in her essay on hacking platform capitalism, points out that cleaning service companies such as SweepSouth claim themselves to be a 'technology company' and 'refer to domestic workers registered on their platform as

“SweepStars” and not as workers’, which allows them to circumvent the labour regulations of the cleaning service or contract service or employment service companies, and denies their ‘contractual identity’ as a worker. The lack of access to any kind of labour protection – unemployment benefits, sick leave, paid leave, social protection – adds to their exploitation. In addition, Ghosh *et al.*, in ‘Locating women workers in the gig economy in India – *old wine in a new bottle?*’, show that due to capital investments that these workers as ‘independent contractors’ have to invest to set up their business, there is a high risk of getting trapped in a cycle of debt as availability of work on these platforms is quite irregular. While platforms do not recognise these workers as their employees, Rodriguez *et al.*, in their study ‘Platform work in the domestic and home care sector: new mechanisms of invisibility and exploitation of women migrant workers’, found that the arrival and departure times of workers are monitored by ‘invisible’ managers without providing them an employee status. Both the flexibility and misclassification of workers work in favour of the platform, as they have a workforce available 24/7 who can be exploited without having to provide any work-related and social security benefits.

The promise of flexibility is often also impinged upon by the control that the platforms exercise over the work schedules and access to work through the rating systems. Sibiya and du Toit, in their paper on domestic worker platforms in South Africa, show that women are pressured to provide quality cleaning services as it has an impact on their ratings and future access to work; while often workers are not given sufficient information about how big the house is and are at times torn between leaving the work unfinished as their time is up (Kalla). As Kalla argues, the focus on time-based tasks can be ‘a double-edged sword’, contractually and algorithmically (p. 659). In addition, workers who cancel or refuse a task, or resist doing extra work, which was not initially specified in their booking, can be penalised through lower ratings (Tandon and Sekharan ; Ghosh *et al.*; Sibiya and du Toit). Ratings are also used to assign workers to ‘different service categories – classic, prime, and lux, which then decides work availability, commission rates, length of commute, and prices of products and equipment’ in beauty work platforms in India (Tandon and Sekharan, p. 693). Further, worker accounts can be deactivated or suspended without any recourse if the ratings fall below a particular threshold or if they repeatedly refuse client bookings in the case of ride-hailing apps (Kwan; Micha *et al.*) and domestic workers platforms (Sibiya and du Toit; Kalla). Women workers’ vulnerability is amplified by this constant fear of penalties, downgrading of ratings, and deactivation.

Women in male-dominated occupations are doubly disadvantaged with the algorithmic rating systems as lower ratings have an impact on their future price of the ride as well as the type of service offered. Micha *et al.*, in ‘When women enter male-dominated territories in the platform economy: gender inequalities among drivers and riders in Argentina’, argue that platforms reward workers with intensive dedication, and when women cancel or refuse a long-distance ride due to care responsibilities then they are penalised with low ratings, paid lower rates, and their participation is also restricted during peak hours. Apart from being penalised by the algorithm, women workers also face pure discrimination from passengers who often cancel the ride because they are women or, when they take the ride, they tell them that ‘women are not safe drivers’ and give them negative

reviews and low ratings, as was observed in China (Kwan). The gender stereotyping of worker capability and performance also has huge implications on women workers' access to work and stable income.

The algorithmic management practices further reinforce and weaken women's position in male-dominated occupations, and they are penalised in terms of their earnings. The use of 'dynamic' or 'surge' pricing during evenings, nights, and weekends, along with bonuses and incentives to ensure a regular supply of workers, discriminates against women. Due to women's care responsibilities and safety reasons they often tend to work less during this period, which puts them in a disadvantageous position. Micha *et al.* find that only 40 per cent of women work during evenings and weekends in Argentina compared to 80 per cent of men. This leads to significant differences in gender gaps in working hours as well as incomes as men are able to benefit from the 'surge' pricing and the various bonuses and incentives that are associated with it (Micha *et al.*; Centeno Maya *et al.*). Barzilay (2019) argues that there is a gender bias in the way algorithms are designed, as it assumes a worker to be a male breadwinner without any care and household responsibilities, and without similar safety concerns. The lack of gender-sensitive or responsive algorithmic systems reproduces and reinforces unequal gender dynamics, precariatizing women workers. Kalla also points out that in addition to unstable work and wages on these platforms, workers also do not receive any employment slips which makes planning for living costs or opening accounts very difficult. A number of papers also highlight the high commission fees that are charged from their earnings by the platforms, which led to some major protests in India, as explained by Dhar and Thuppilikkat in 'Gendered labour's positions of vulnerabilities in digital labour platforms and strategies of resistance: a case study of women workers' struggle in Urban Company, New Delhi' and Tandon and Sekharan, and is also contrary to the International Labour Standards (ILO Protection of Wages Convention, 1949 (No. 95) and ILO Private Employment Agencies Convention, 1997 (No. 181)) which prohibits the practice of charging fees to workers.

While platforms give an opportunity to women to enter male-dominated sectors, and an 'immense sense of self-confidence and self-esteem' (Ghosh *et al.*, p. 780) and altering the gender segregation in occupations (Micha *et al.*), the reality of women workers seems to be quite different. The lack of decent wages and the dependence on the rating-based algorithm makes work unstable and insecure, and many women are frustrated as they do not find enough clients on platforms and must look for alternative short-term jobs (Sibiya and du Toit).

Further, penalties and rewards also force workers to work long working hours often beyond their physical and mental capacities with implications on their health, which is highlighted in several papers (Ghosh *et al.*; Kwan; Tandon and Sekharan). Despite these injustices, women workers are quite instrumental in navigating opaque and inefficient algorithms. Tandon and Sekharan show that workers on beauty platforms tried to regain some control over working hours and earnings by identifying the inefficiencies in the algorithmic task assignment and reworking it to their advantage by utilising social networks and peer groups and adopting resilience strategies.

Many of the papers bring out how women workers navigate between paid work, unpaid work, and care responsibilities, and the false promise of the flexibility of platforms. The papers also bring out that the precarious working conditions on the platforms and increases in women's both paid and unpaid work burdens might not be beneficial for women's well-being. Kwan highlights how many vulnerable women, like single mothers, take risks to earn money or build their savings, such as working during illness and menstruation, or while exhausted, which has implications on their health. In addition, the nature of work, whether ride-hailing (male-dominated that entails travelling long distances at night) or beauty work and domestic work (which entail work conducted within the privacy of clients' homes), all carry unique risks to women workers, with exposure to sexual harassment and violence being prevalent in all platforms. Women often face a dilemma between risking their safety by working at night or being penalised through algorithmic ratings if they cancel or reject work (Dhar and Thuppilikkat). Kwan's paper highlights gender-based violence, including sexual harassment and unsafe rides at late hours of the night with no safe and accountable systems provided by the platform. Centeno Maya *et al.*, in their study of food delivery workers in Mexico City, show that women deal with such risks by having someone accompany them to reduce insecurity (harassment, theft, fear of being kidnapped or raped, insecurity in streets and road safety) and discrimination. This precarious situation on the platforms further deepens the existing gendered intersectional inequalities, making women workers' positions more precarious, insecure, and vulnerable.

Digital platforms also take advantage of inequalities of gender, race, and immigration status to access a precarious workforce. Rodriguez *et al.* in their paper describe the new mechanisms of exploitation and invisibility that women face in domestic and home care work, which are mainly carried out by migrant women from the global South. The low reservation wage and lack of agency of migrant women, who are denied access to other sources of income and formal employment, act as key elements in the advancement of the mechanisms of exploitation and exclusion. Though care platforms facilitate access to work for migrant women, their working conditions are characterised by precarity, lack of access to social protection, and unemployment benefits. They argue that digital platforms have reinforced the 'casualisation' of labour markets, as well as gendered segregation and subjugation in labour markets. As Dixit and Banday illustrate, we are witnessing a different form of global domination and the reproduction of colonial frameworks of development that advance neoliberal solutions and further marginalise women in the 'third world'.

Many contributions in this special issue also question the 'empowering' potential of digital labour platforms on women, particularly those belonging to marginalised groups, by offering an intersectional analysis of the gender and racial dynamics. In addition, there has been a systematic lack of acknowledgement by labour platforms of the gendered labour of social reproduction, which has led to further informalisation of work and precaritisation of marginalised women, a concern that is highlighted by a number of contributions in this issue. These are manifested in low-paid task-based, fragmented work, charging high commission rates, lack of social protection, lack of autonomy over

work, absence of regulatory frameworks to protect workers, and lack of health and safety measures or accountability.

Anwar, based on interviews with women working on both remote and place-based platforms in Ghana, Nigeria, Uganda, Kenya, and South Africa, highlights the physical, financial, and emotional repercussions that women working on these platforms face and further reiterates the argument that the mere inclusion and presence of women in digital platforms are not going to advance their empowerment, and he argues that the platforms have in fact in many ways further re-entrenched gender and race-based inequalities and discriminations.

Ghosh *et al.*, in this issue, unpack the highly gendered and precarious nature of platform work in the feminised sector of beauty work and the male-dominated ride-hailing platforms. They assert that while women in both sectors are able to express their agency to an extent like choosing the sector, upskilling, and being mobile, overall, the ability to exercise choice in their personal and professional lives is limited by prevailing gender norms in digital labour platforms, which hinders their ability to achieve empowerment. Moreover, the inability to collectivise and unionise severely impacts their agency and quality of life. The rise of labour platforms in a number of service industries is the continuation of the informal nature of work leading to increasing insecurity and they conclude that this form of work is nothing but ‘old wine in a new bottle’ (p. 781).

Digital labour platforms have thus changed the way work is organised and the labour relations of millions of workers across the world. It is shaping work processes and relations as well as the power dynamics that structure the work. The platformisation of work and the increased use of algorithmic management for organising and monitoring work, has indeed led to the informalisation of employment relations and the blurring of boundaries between paid and unpaid care work (Sun *et al.* 2021). Platformisation of work, as shown through this issue, is resulting in greater informalisation, commodification, fragmentation, and precariatization of work and labour.

Worker protests and emerging forms of worker organisation

As discussed, digital labour platforms have exacerbated informality by misclassifying workers as independent contractors or self-employed or partners, and managing workers through algorithmic management practices. If, on the one hand, trade unions have struggled to respond to the introduction of labour-saving technology in historically unionised sectors, on the other hand, they have been reluctant to organise platform workers, due to the absence of a clear employment relationship. Nevertheless, platform workers have sought to resist and reshape the terms of platform capitalism, both individually and collectively, as workers and as women. Ghosh *et al.*, in their paper, describe how women on delivery platforms have adopted self-defence tactics rather than relying on platforms to create safe workplaces. Sibiyi and du Toit describe how domestic workers in South Africa used social media platforms like Facebook and WhatsApp to share information and opportunities, request assistance, vent their frustrations, and reclaim a sense of dignity – a phenomenon also highlighted by Dhar and Thuppilikkat.

In China, to address the gender-based violence in male-dominated platforms, women platform workers explicitly excluded men and platform managers from their social media groups. Kwan explains that given the gendered and sexualised work environment, women drivers preferred women-only WeChat and TikTok groups to communicate their experiences and build solidarities, and seek support while on the road. She shows that these WeChat groups play a significant role in providing a safe space ‘for women drivers to address the confrontations they face in a masculine work environment’ (p. 564).

In some instances, these forms of co-ordinated individual resistance have coalesced into collective action. Tandon and Sekharan explore how WhatsApp groups have also been used to co-ordinate strike action. They argue that women’s networks of information sharing and care are instrumental in navigating opaque and inefficient algorithms. While women workers self-identify as partners or entrepreneurs, their demands include fair and transparent models of partnership with lower commission rates, equitable allocations of work, proper grievance procedures, workplace safety, and time to balance care work. Their sense of identity has been instrumental to collective organising and has drawn on a rich history of organising in the informal economy.

In some countries, women platform workers have developed co-operatives. Salvagni *et al.*, in ‘Gendering platform co-operativism: the rise of women-owned rider co-operatives in Brazil and Spain’, argue that in Brazil co-operatives offer an opportunity to advance a more inclusive, feminist digital economy, which supports women in work and motherhood. Though membership numbers may still be relatively small, they offer alternative visions at the intersection of work and technology, and therein lies their transformative power. Rather than try to compete with Big Tech food delivery platforms, they have developed alternative business models which emphasise environmental sustainability, quality products, and decent work. Platform co-operatives as an alternative form of organisation is also echoed by Kwan, as it has the potential to enable more equitable labour outcomes. However, gender platform co-operativism supportive public policies are key. Furthermore, some of the existing co-operatives have reoriented themselves and Agarwal and Chatterjee, in ‘Leveraging digital technologies to enable women’s co-operatives: experience of national insurance VimoSEWA Co-operative Ltd.’, discuss how VimoSEWA, a micro-insurance entity serving informal workers since 1992 across several states in India, leveraged digital technologies in its entire operation to ensure that women from marginalised groups continued to be financially covered throughout the COVID-19 pandemic.

In other countries, platform workers have established trade unions, whether formally or informally. Dhar and Thuppilikkat illustrate how women have drawn on information and communication technology, as well as informal kin networks, to co-ordinate protest actions against digital labour platforms, with the support of the established trade union movement. Their success, albeit limited, points to their associational and social power. Despite budding victories, Kwan argues that public policies are key in defining more dignified and decent business models for digital labour platforms.

The role of the state and public policies

Given the rapid pace of digitalisation, governments have struggled to regulate digital labour platforms (ILO 2021). As platforms have classified workers as independent contractors or self-employed, they are absolved from providing labour and social protections. Centeno Maya *et al.*, in their paper on food delivery workers in Mexico City, note that ‘The economic power of some of the corporations behind the apps has grown so much, and the pandemic has made their services so useful, that it is becoming increasingly hard to leverage enough political power to implement serious regulation’ (p. 615). However, it is important to recognise that classification of workers as employees does not automatically address poor working conditions and it would be important to ensure that workers, irrespective of whether they are classified as self-employed or employees, should enjoy the right to associate, bargain collectively, and be provided with universal labour and social protections. Furthermore, while the state has an important regulatory role to play, the process of formalisation requires the inclusion and active participation of platform workers.

In this context, digital technologies can also be effectively harnessed to provide decent work to all workers. For instance, in several countries governments use digital technologies to promote formalisation through digital payments, providing social protection and other benefits, filing and payment of taxes, among others (Chacaltana *et al.* 2018). This strategy could be used for platform work. Sibiya and du Toit discuss ways in which digital labour platforms could be used to professionalise domestic work and to offer them more than minimum wage, while at the same time regulating the number of hours of work.

The state can also play an important promotive role, whether through reskilling and upskilling women workers (von Dietrich and Garcia), supporting the creation of women-owned platform co-operatives (Salvagni *et al.*), or improving access to digital technologies (Majid and Mustafa). Salvagni *et al.* discuss a 2021 initiative in Barcelona, for instance, where ‘the Open University of Catalonia and the City of Barcelona developed a partnership with three pillars: training, orienting and financing cross-sectoral platform co-operatives with a gender focus; strategic collaborations to ensure gender diversity in technology companies and university STEM research; and the building of platform prototypes’ (p. 719). State support is especially important given the highly competitive nature of digital labour platforms. A more inclusive digital economy also requires investment in research and development; infrastructure, including digital infrastructure; public services, such as education, health, child care, and safety; and social protection, including both social insurance and social assistance (von Dietrich and Garcia; Micha *et al.*). Ultimately, the context matters, and it is important to understand the barriers to digital inclusion in order to introduce appropriate public policies (Majid and Mustafa).

It is important to keep in mind that public policies are a site of contestation precisely because they touch on the question of power. Indeed, policymaking processes are often shaped by dominant ideological frames, and underlying biases and struggles between competing interests. Therefore, it is critical that women and other marginalised groups be actively involved in shaping the terms of public policies. Raj and Juned, in their

paper on the lived realities of transgender people, show how with digitalisation they are increasingly alienated from public provisioning and are subjected to ‘excessive requirements and arbitrary rules’, wherein the ‘power of identity determination and issuance of ID [identity] cards are placed on intermediaries such as lawyers who issue affidavits, or the DM [District Magistrates] Office which issues the cards and certificates, rather than individuals themselves’ (p. 540). As a result, ‘at the ground level trans persons are forced to interact with government officials and lawyers and witness administrative apathy and social harassment’ (p. 539).

Furthermore, it is critical to interrogate the narrative frames which have risen to dominance and increasingly shape policymaking processes. Dixit and Banday argue that the dominant discourse on bridging the gender divide in digital technologies, promoted by international development agencies, ultimately reproduce colonial and neoliberal discourses. While platform work is widely decried as precarious employment in the global North, it has been punted as a source of entrepreneurship in the global South (Anwar). Ultimately, public policies must be grounded in a feminist and decolonial analysis of gender, race or caste, class, and other dynamics which structure society. The inclusion of workers, women, and other marginalised populations in designing, and monitoring policies is thus paramount (Raj and Juned). Overall, the contributions in this special issue advocate for a more contextualised, participatory, and accountable process of policy formulation to ensure that digital policies are grounded in local, social, and economic contexts, that they recognise the structural/sociocultural norms that shape labour markets and economic participation, and that they reflect the voices of those who have been historically marginalised and excluded from technological advancements including digitalisation.

Conclusion

Technological innovation has always been an inherent feature of capitalism, driven by capital’s desire to expand its frontiers of accumulation in periods of stagnation, through a ‘technological fix’. The long-term impact of new technologies on labour markets and conditions of work is still unclear. Although projections of labour replacement show cause for concern, there are many mediating factors which will shape the rollout of new technologies in the global South, including levels of investment and the nature of social dialogue. After all, the process of technological innovation is neither linear nor predetermined. Indeed, as Silver (2014) points out in her historical analysis of the making, unmaking, and remaking of working classes, where capital flows, labour conflict soon follows.

Having said that, new technologies have facilitated growing labour insecurity through the proliferation of digital labour platforms. Most workers on labour platforms are misclassified as self-employed, independent contractors, or partners rather than employees, absolving employers from having to comply with regulations regarding labour and social protections. Platforms argue that self-employment offers greater flexibility – and indeed, the promise of flexibility is one of the reasons that workers are drawn to digital labour

platforms. However, the combination of algorithmic management and a task-based system of remuneration based on the tyranny of low wages means that platforms provide very little flexibility at all. Consequently, workers work long hours under precarious conditions, deepening inequality along the lines of gender, race, caste, class, and geography.

What gives digital labour platforms power is the absence of regulations governing them as they are self-governed. Therefore, the regulation of digital labour platforms and the introduction of public policies to support an emancipatory form of digital work are key. Given the importance of data to algorithmic management on platforms – from allocation of work, monitoring, to ratings and remuneration, to deactivation – and to processes of accumulation, it is critical to determine the ownership over the data collected. However, regulation is itself an exercise of power and thus workers must be included in regulatory efforts. While the collective organisation of platform work is still limited, it is on the rise through social media platforms, informal worker organisations, co-operatives, and trade unions. The questions, therefore, are: How to regulate digital labour platforms without undermining worker autonomy?; How to build worker organisations at the intersections of various lines of exploitation, exclusion, and oppression?; and How to forge alliances between emerging forms of worker organisation and established trade unions?

Ultimately, what is produced, how it is produced, and for whom is part of a political process, the outcome of which reflects the social forces at play. While new technologies are deepening inequalities, it is not the technology *per se* which is the problem, but how it is used. As Anner *et al.* (2019, 3) point out: ‘When workers are organised and given a voice in processes of work transformation, and when governments actively support that process of worker engagement, then technology has the potential to play a positive role in the future of work.’

The papers in this special issue are an important contribution to the growing body of literature on new technologies, the future of work, and implications for inequality. By focusing on women workers and women-dominated sectors they offer important insights into the gendered nature of platform work. However, they also highlight the gendered forms of worker organisation, mobilisation, and resistance, as workers navigate a male-dominated industry. These range from everyday strategies to access digital technologies to the formation of trade unions, and everything in between. By employing an intersectional lens, they explore the struggles and contestations at the intersection of various forms of exploitation and oppression.

As with any special issue, this issue raises new questions for analysis. The first is the need to locate the emergence of platform work within historically specific processes of capitalist development, structural transformation, and labour market restructuring. The rise of digital labour platforms is both a reflection of and a response to variegated forms of neoliberalism; and it would be important for an emancipatory project, how these are intertwined with broader processes of accumulation with the commodification of social reproduction, financialisation of production processes, and securitisation.

A second question is how to conceptualise the diverse array of digital labour platforms and emerging sources of worker power or contestation. As the papers in this special issue

illustrate, platforms encompass numerous sectors, each with its specificities. Furthermore, platforms are not necessarily homogenous; while some impose high levels of algorithmic management, others impose far less. Moreover, workers have plural identities and aspirations which may not be fully actualised by standard employment relations. Therefore, it is critical to develop different typologies which look at the nature of employment relations, worker aspirations, and the sources of worker power. However, whatever typology is developed, it would be imperative to ensure that they have both labour and social protection.

A third and final question is how to reimagine regulation at the intersection of data, labour, and social protections. As worker struggles expand and regulatory reforms are tested, it is important to continue to collect case studies of the successful organisation and regulation of digital labour platforms, including data and algorithms.

Notes on contributors

Uma Rani is Senior Economist at the Research Department of the International Labour Office, Geneva. Her research interests include informal economy, minimum wages, and social policies. Since 2016, her research has focused on transformations in the digital economy, wherein she explores how labour and social institutions interact with public policies and can provide decent working conditions to workers. Email: amara@ilo.org

Ruth Castel-Branco is a Senior Researcher and Manager of the Future of Work(ers) research group at the Southern Centre for Inequality Studies at the University of the Witwatersrand. She holds a PhD in Sociology from the University of the Witwatersrand (2021). Her interests include structural transformation and the changing nature of work, workers' rights and organisation, the redistributive role of the state, and social protection. Prior to pursuing a PhD, she worked on maternity and social protection at the International Labour Office in Mozambique. Email: Ruth.Castel-Branco@wits.ac.za

Shivani Satija is the Editor of *Gender & Development*. She holds a PhD in Women's Studies from the Tata Institute of Social Sciences, Mumbai. Her research interests include gender-based violence, caste, labour, vulnerability and resistance, multi-modal research, and critical online practice. Her recent research examines the potential of online writings by marginalised groups in rethinking bodily vulnerability and shaping popular digital culture along principles of inclusion, plurality, and gender justice. Email: shivani.satija@oxfam.org

Mahima Nayar is a trained psychiatric social worker and an independent researcher working in the areas of disability, mental health, women, children, and families. She worked as Assistant Professor at the Tata Institute of Social Sciences, Mumbai. She has also worked extensively with survivors of trafficking (women and children), women facing domestic violence and survivors of sexual assault, persons with psychosocial disabilities, and their families. She is the author of the book *Against All Odds: Psychosocial Distress and Healing Among Women* (Sage Yoda 2018) and co-editor of the book *The Contemporary Indian Family: Transitions and Diversity* (Routledge 2020). Email: mahimanayar@gmail.com

ORCID

Uma Rani  <https://orcid.org/0000-0002-4874-8689>

Ruth Caste-Branco  <https://orcid.org/0000-0001-9907-2503>

Shivani Satija  <http://orcid.org/0000-0001-7206-0166>

Mahima Nayar  <http://orcid.org/0000-0002-3088-3903>

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